In this lab, you'll learn how to deploy your application to AWS using Infrastructure as Code (IaC). You can choose between two popular IaC tools: AWS Cloud Development Kit (CDK) or Terraform. Both will create a deployment that uses Amazon S3 for hosting your static files and Amazon CloudFront for content delivery.

Prerequisites

> **Note**
> Complete Lab 1 - Setting Up Kiro

Choose Your IaC Tool

Before deploying your application, choose between two Infrastructure as Code (IaC) approaches:

Option A: AWS Cloud Development Kit (CDK)

Best for:

- AWS-focused projects
- Developers comfortable with Python, TypeScript, Java, or C#
- Teams wanting AWS-native abstractions and constructs
- Projects requiring tight AWS service integration

Advantages:

- High-level constructs for common AWS patterns
- Strong type safety and IDE support
- Automatic CloudFormation template generation
- Rich ecosystem of AWS constructs

Follow these steps:

1. Option 1: Deploy with AWS CDK

Option B: Terraform

Best for:

- Multi-cloud or cloud-agnostic projects
- Teams familiar with HashiCorp Configuration Language (HCL)
- Organizations with existing Terraform expertise
- Projects requiring fine-grained resource control

Advantages:

- Works across multiple cloud providers
- Declarative configuration language
- Mature ecosystem and community
- Explicit resource management

Follow these steps:

1. Option 2: Deploy with Terraform

> **Note**
> Both approaches will create the same AWS infrastructure:
>
> - Amazon S3 bucket for hosting static files
> - Amazon CloudFront distribution for content delivery
> - Proper security configurations and permissions

Choose one path and proceed with the corresponding steps.

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